

QBUS2820 Predictive Analytics, S2 2026

Week 2: Linear Regression

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Simple Linear Regression

Multiple Linear Regression

Recommended reading

- Chapter 3, *An Introduction to Statistical Learning with Applications in R* by James et al.: easy to read, sloppy discussions sometimes, comes with R/Python code for practice.
- Chapters 3, *The Elements of Statistical Learning* by Hastie et al.: well-written, deep in theory, suitable for students with a sound maths background.

Spending Data

The Spending Data contains information on 1000 customers in a customer database for the company Direct Marketing (DM). The data consist of

- AmountSpent (\$): the amount spent by each customer in one year on DM products.
- Salary (\$)
- Catalogs: number of shopping catalogs sent to each customer per year
- Children: number of children each customer has
- and many others

The company would like to build a model to explain/predict the amount each customer spending on its products, based on the other variables.

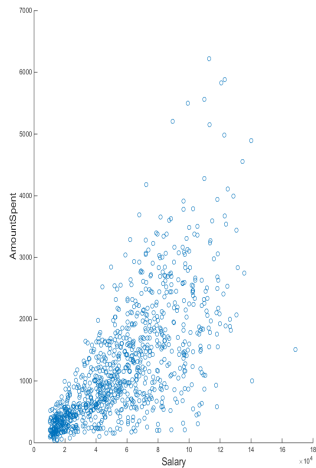
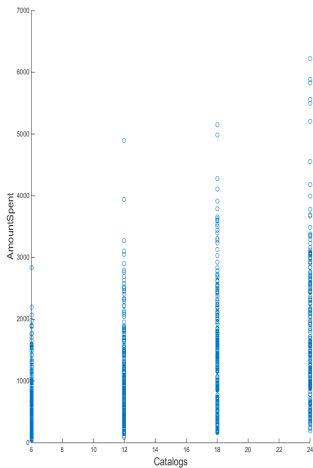
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Spending Data



Spending Data

Questions might be asked

- Given a particular customer, how much **spending amount** would the company expect from he/she?
- Is there a **relationship** between the spending amount and the number of catalogs, for example?
- How does the salary affect the spending amount?
- How do the salary and the number of children **interactively** affect the spending amount?
- What is the **best subset** of the explanatory variables in terms of predicting the spending amount?
- ...

Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

- Y : the **response variable** or **dependent variable** to be predicted
- $X_1, \dots, X_p$: potential **predictors** or **covariates** or **independent variables**
- the intercept β_0 and slope coefficients $\beta_1, \dots, \beta_p$ are **unknown population parameters** to be estimated
- ϵ : error term or something that can't be explained by the model. Assume $\mathbb{E}(\epsilon) = 0$ and $\mathbb{V}(\epsilon) = \sigma^2$, σ^2 is also unknown.

Caution: the model imposes two main assumptions that need to be checked/testified! Y should be continuous, **not suitable** for categorical/binary response variables. The X_j can be both continuous/discrete or categorical.

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Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

- What does 'linear' mean? Linear in X or β ?
- The model is linear in the sense that the coefficients $\beta_0, \beta_1, \dots, \beta_p$ are combined linearly with the independent variables.
- E.g.,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 e^{X_2} + \beta_3 X_1^3 + \beta_4 X_1 X_2 + \epsilon$$

is still a linear regression model in terms of $Z_1 = X_1$, $Z_2 = e^{X_2}$, $Z_3 = X_1^3$ and $Z_4 = X_1 X_2$.

- Precisely, X_1, X_2 are called **covariates**. In this model, X_1 , X_1^3 , e^{X_2} and $X_1 X_2$ are called **predictors** or **features**

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$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

- Why do we start from linear regression?
- Linear regression might sound too simplistic, it's hardly true that Y is linearly dependent on the X_j 's
- But linear regression is extremely useful and important. It **forms a basic framework** for non-linear regression and many advanced regression models, such as **regularized regression**, **support vector machines (SVMs)** and **deep neural networks (DNNs)**...

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Simple Linear Regression

Simple Linear Regression

- Given a quantitative output variable Y and an input variable X . The task is to explain/predict Y given a value of X .
- Fix a particular value x of X
- Recall from Lecture 1: Given the squared loss function, the optimal prediction of Y when $X = x$ is $f(x) = E(Y|X = x)$
- In general, the function $f(x) = E(Y|X = x)$ is unknown.
- Key assumption in Linear Regression: we assume that

$$f(x) = E(Y|X = x) = \beta_0 + \beta_1 x$$

where β_0 and β_1 are some parameters.

Simple Linear Regression

This leads to the following linear regression model:

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \mathbb{E}(\epsilon) = 0, \quad \mathbb{V}(\epsilon) = \sigma^2$$

So the conditional mean of Y given $X = x$ is a linear function of x

$$\mathbb{E}(Y|X = x) = \beta_0 + \beta_1 x$$

Our working example

$$\text{AmountSpent} = \beta_0 + \beta_1 \text{Catalogs} + \epsilon$$

Simple Linear Regression

Our working example

$$\text{AmountSpent} = \beta_0 + \beta_1 \text{Catalogs} + \epsilon$$

Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be estimates of β_0 and β_1 respectively. Then

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$$

is a **point estimate** of $\mathbb{E}(Y|X = x)$. This is an **estimate** of the average spending amount among **all** customers who are sent x catalogs per year.

We also use $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ to **predict** the spending amount of an **individual** customer who is sent x catalogs.

Estimation methods

Estimation methods:

- **Least squares** method;
- **Maximum likelihood** method.

Estimating the coefficients by the least squares method

- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ is the prediction of the observation y_i when $X = x_i$. So $e_i = y_i - \hat{y}_i$ represents the i th residual
- We define the residual sum of squares (RSS) as

$$RSS(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right)^2.$$

Note that $\{y_i, x_i, i = 1, \dots, n\}$ is the training dataset.

- The best $\hat{\beta}_0$ and $\hat{\beta}_1$ will be the ones that minimise $RSS(\hat{\beta}_0, \hat{\beta}_1)$.
- This is known as Least Squares Method. No assumption on probability distribution of ϵ is needed.

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Estimating the coefficients by the least squares method

- $\hat{\beta}_0 = \dots$
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Estimating the coefficients by the least squares method

- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$

Estimation by the maximum likelihood method

- Assume that $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, $i = 1, \dots, n$. Note that

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Therefore $y_i \sim \mathcal{N}(\beta_0 + \beta_1 x_i, \sigma^2)$.

- E.g., the amounts spent by customers who are sent x_i catalogs are normally distributed with mean $\beta_0 + \beta_1 x_i$ and variance σ^2 . Seems a reasonable assumption!
- The **likelihood function** is

$$p(\mathbf{y}|\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right)$$

- Maximising this likelihood with respect to β_0 and β_1 leads exactly to the same least squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$
- Estimate of σ^2 : $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)\right)^2$.

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Brief introduction to Maximum Likelihood Estimation

Given data $\mathbf{y} = \{y_1, \dots, y_n\}$. In almost all areas of applications, we then assume the data come from a **statistical model**, depending on a vector of unknown parameters θ . This model allows us to write down the density $p(y_i|\theta)$ of y_i .

Example. $y_i \stackrel{iid}{\sim} \mathcal{N}(\mu, 1)$, $i = 1, 2, \dots, n$. The joint density of $\mathbf{y}$ is

$$\begin{aligned} p(\mathbf{y}|\mu) &= \prod_{i=1}^n p(y_i|\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{(y_i - \mu)^2}{2}} \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^n \exp\left[-\frac{1}{2} \sum_{i=1}^n (y_i - \mu)^2\right] \end{aligned}$$

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Brief introduction to Maximum Likelihood Estimation

Let $\mathbf{y} = \{y_1, \dots, y_n\}$ be a random sample from a distribution with pdf $p(y|\theta)$. The **likelihood function**, as a function of θ , is defined as

$$p(\mathbf{y}|\theta) = \prod_{i=1}^n p(y_i|\theta).$$

- This likelihood function reflects the probability of observing the data $\mathbf{y}$ if θ is the true parameter
- We wish to estimate the (unknown) true value of θ that generated the data $\mathbf{y}$ by those values that maximise $p(\mathbf{y}|\theta)$.
- The **maximum likelihood estimator (MLE)** of θ is a value that maximises $p(\mathbf{y}|\theta)$.
- MLE is one of the most popular estimation methods. Taught in detail in QBUS3830 Advanced Analytics

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Spending data example

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-15.318	45.374	-0.33759	0.73574
Salary	0.021961	0.00071001	30.93	7.4954e-148

Number of observations: 1000, Error degrees of freedom: 998

Root Mean Squared Error: 687

R-squared: 0.489, Adjusted R-Squared 0.489

F-statistic vs. constant model: 957, p-value = 7.5e-148

Properties of the estimators*¹

Under the assumption that $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, the distribution of $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)'$ is multivariate normal with

$$\text{mean } \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \text{ and covariance } \frac{\sigma^2}{n} \begin{pmatrix} \overline{x^2}/s_x^2 & -\bar{x}/s_x^2 \\ -\bar{x}/s_x^2 & 1/s_x^2 \end{pmatrix}$$

where

$$\bar{x} = \frac{1}{n} \sum x_i, \quad \overline{x^2} = \frac{1}{n} \sum x_i^2, \quad s_x^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \overline{x^2} - (\bar{x})^2$$

That is

$$\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \sigma^2 \frac{\overline{x^2}}{ns_x^2}\right), \quad \hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{ns_x^2}\right)$$

The **standard errors** of the estimators are

$$se(\hat{\beta}_0) = \frac{\sigma}{s_x} \sqrt{\frac{\overline{x^2}}{n}}, \quad se(\hat{\beta}_1) = \frac{\sigma}{\sqrt{n}s_x}$$

¹Technical slide, not tested in the exam

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Uncertainty of the estimates

σ is estimated by

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

So the standard errors of $\hat{\beta}_0$ and $\hat{\beta}_1$ are estimated by

$$\widehat{se(\hat{\beta}_0)} = \frac{\hat{\sigma}}{s_x} \sqrt{\frac{x^2}{n}}, \quad \widehat{se(\hat{\beta}_1)} = \frac{\hat{\sigma}}{\sqrt{n}s_x}$$

The 100%(1- α) confidence interval for β_j is

$$\hat{\beta}_j \pm t_{n-2, \alpha/2} \widehat{se(\hat{\beta}_j)}, \quad j = 0, 1.$$

Here $t_{\nu, p}$ denotes the number such that

$$\mathbb{P}(t_{\nu} > t_{\nu, p}) = p, \quad p \in [0, 1]$$

where t_{ν} is a random variable with Student's t distribution with ν degrees of freedom.

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Testing about the population coefficients

Question of interest: is there a relationship between Salary and AmountSpent?

$$H_0 : \beta_1 = 0 \text{ v.s. } H_1 : \beta_1 \neq 0$$

- Test statistic

$$t_{\text{stat}} = \frac{\hat{\beta}_1 - 0}{\text{se}(\hat{\beta}_1)}$$

- Under the null hypothesis H_0 , $t_{\text{stat}} \sim t_{n-2}$, i.e. it follows the Student's t distribution with $n - 2$ degrees of freedom.
- Statistical software often reports **p-value**

$$p\text{-value} = \mathbb{P}(t_{n-2} > |t_{\text{stat}}|)$$

- The smaller this **p-value**, the more evidence is **against** H_0 .
Typical cutoff p-value: 0.05 or 0.01.

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Spending data example

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-15.318	45.374	-0.33759	0.73574
Salary	0.021961	0.00071001	30.93	7.4954e-148

Number of observations: 1000, Error degrees of freedom: 998

Root Mean Squared Error: 687

R-squared: 0.489, Adjusted R-Squared 0.489

F-statistic vs. constant model: 957, p-value = 7.5e-148

Similarly, we can test $H_0 : \beta_0 = 0$ v.s. $H_1 : \beta_0 \neq 0$. But this is not statistically interesting!

Multiple Linear Regression

Multiple linear regression

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i = x_i' \beta + \epsilon_i \quad i = 1, \dots, n$$

$$\mathbb{E}(\epsilon_i) = 0, \quad \mathbb{V}(\epsilon_i) = \sigma^2, \quad \beta = (\beta_0, \dots, \beta_p)' \in \mathbb{R}^{p+1}.$$

$$X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{i1} & \cdots & x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_n \end{bmatrix}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_i \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

Matrix form

$$\mathbf{y} = X\beta + \boldsymbol{\epsilon}$$

The least squares method estimates β by minimising the RSS

$$RSS = \sum_{i=1}^n (y_i - x_i' \beta)^2 = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta).$$

Least squares estimate^{*2}

$$\frac{\partial RSS}{\partial \beta} = -2X'(\mathbf{y} - X\beta)$$

The solution is

$$\hat{\beta}_{ls} = (X'X)^{-1}X'\mathbf{y},$$

given that the inverse of matrix $X'X$ exists.

Because

$$\frac{\partial^2 RSS}{\partial \beta \partial \beta'} = 2X'X$$

is positive definite, RSS is minimised at $\beta = \hat{\beta}_{ls}$.

²Technical slide, not tested in the exam

Multiple linear regression

If we assume a normal distribution for the error terms, i.e. $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$, then $y_i \sim \mathcal{N}(x_i' \beta, \sigma^2)$.

The density of y_i given x_i and parameters $\theta = (\beta', \sigma^2)'$

$$p(y_i|x_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - x_i'\beta)^2\right).$$

The log-likelihood

$$\ell(\theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - X\beta)'(\mathbf{y} - X\beta)$$

The **maximum likelihood estimator (MLE)** of β

$$\hat{\beta} = (X'X)^{-1}X'\mathbf{y} = \hat{\beta}_{ls}.$$

MLE of σ^2 is

$$\hat{\sigma}^2 = \frac{(\mathbf{y} - X\hat{\beta})'(\mathbf{y} - X\hat{\beta})}{n}$$

Note: In some textbooks the denominator n is replaced by $n-p-1$ so that $\mathbb{E}(\hat{\sigma}^2) = \sigma^2$.

Multiple linear regression

-

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E}[(X'X)^{-1}X'y] = \mathbb{E}[(X'X)^{-1}X'(X\beta + \epsilon)] \\ &= \beta + (X'X)^{-1}X'\mathbb{E}(\epsilon) = \beta\end{aligned}$$

So $\hat{\beta}$ is unbiased estimator of β .

- If $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X'X)^{-1})$.

These properties of the estimators are important for constructing confidence intervals, hypothesis testing.

Multiple linear regression

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These properties of the estimators are important for constructing confidence intervals, hypothesis testing.

Testing about the population coefficients

Question of interest: is there a relationship between Salary and AmountSpent?

$$H_0 : \beta_j = 0 \text{ v.s. } H_1 : \beta_j \neq 0$$

- Test statistic

$$t_{\text{stat}} = \frac{\hat{\beta}_j - 0}{\text{se}(\hat{\beta}_j)}$$

- Under the null hypothesis H_0 , $t_{\text{stat}} \sim t_{n-p-1}$, i.e. it follows the Student's t distribution with $n - p - 1$ degrees of freedom.
- Statistical software often reports *p-value*

$$p\text{-value} = \mathbb{P}(t_{n-p-1} > |t_{\text{stat}}|)$$

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Spending data example

$$\text{AmountSpent} = \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Children} + \beta_3 \times \text{Catalogs} + \epsilon$$

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-442.76	53.718	-8.2423	5.2872e-16
Salary	0.020406	0.0005929	34.417	1.1637e-171
Children	-198.69	17.087	-11.628	2.1322e-29
Catalogs	47.695	2.7553	17.31	6.8068e-59

Number of observations: 1000, Error degrees of freedom: 996

Root Mean Squared Error: 563

R-squared: 0.658, Adjusted R-Squared 0.657

F-statistic vs. constant model: 640, p-value = 9.56e-232

Are Salary, Children, Catalogs and the intercept significant?

Spending data example: interpreting the coefficients

- Among all the customers who have the same number of children and receive the same number of catalogs, on average, a customer spends $\hat{\beta}_{\text{salary}} = 2.04$ cent from each \$1 increased in salary
- Fix Salary and Catalogs, on average, AmountSpent is **increased** by $\hat{\beta}_{\text{children}} = -198.69$ \$ on each 1 unit increased in Children.

That is, among all customers who have the same salary and receive the same number of catalogs, on average, a customer spends \$198.69 **less** when having one more child.

- Similarly, given the same salary and number of children, on average, AmountSpent is increased by \$47.695 when receiving one more catalog per year.

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Is the entire linear regression model significant at all?

We may ask: is the linear regression model significant in terms of explaining Y ? In other word, is the set of predictors $X_1, \dots, X_p$ useful at all for explaining Y ?

$H_0 : \beta_1 = \dots = \beta_p = 0$ v.s. H_1 : at least one $\beta_j \neq 0$

- Test statistic

$$F_{\text{stat}} = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}$$

where $TSS = \sum (y_i - \bar{y})^2$ is the **total sum of squares**

- Under H_0 , F_{stat} has a F -distribution with degrees of freedom p and $n - p - 1$.

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Not all predictors are important

- Usually, the number of potential predictors p is large. Not all variables $X_1, \dots, X_p$ are significant for predicting Y
- One needs to perform the variable selection task: select a subset of these p variables that is the best for predicting Y
- One approach:
 - starting with the full model
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 - Repeat until all coefficients are significant
- We will discuss variable selection and model selection in much more detail later in the course

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Dealing with categorical covariates

- Some covariates might not be **quantitative**, but **qualitative** or **categorical**. E.g, gender, marital status
- We use **dummy variables** to represent qualitative covariates
- In the spending data, Gender_b is a dummy variable that represents gender of customers

$$\text{Gender_b} = \begin{cases} 1, & \text{if the customer is female} \\ 0, & \text{if the customer is male} \end{cases}$$

Spending data example

$$\text{AmountSpent} = \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Children} + \beta_3 \times \text{Catalogs} + \beta_4 \times \text{Gender_b} + \epsilon$$

	Estimate	SE
(Intercept)	-472.99	60.071
Salary	0.020585	0.00061386
Children	-200.99	17.207
Gender_b	41.749	37.152
Catalogs	47.778	2.7559

For female customers

$$\text{AmountSpent} = \beta_0 + \beta_4 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Children} + \epsilon$$

For male customers

$$\text{AmountSpent} = \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Children} + \epsilon$$

All else being equal, a female customer is spending on average $\hat{\beta}_4 = \$41.749$ more than a male customer.

Dealing with categorical covariates

In general, we need $K - 1$ dummy variables to represent a qualitative variable with K categories.

A categorical variable C with K categories/levels $c_1, \dots, c_K$ is represented by $K - 1$ variables $X_1, \dots, X_{K-1}$

$$X_1 = \begin{cases} 1, & \text{if level } c_1 \\ 0, & \text{else} \end{cases}, \dots, X_{K-1} = \begin{cases} 1, & \text{if level } c_{K-1} \\ 0, & \text{else} \end{cases}.$$

Level c_K is **baseline** level: all other levels are compared against this baseline level.

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Spending data example

Age has three categories: Young, Middle or Old.

$$\text{Age}_y = \begin{cases} 1, & \text{if young} \\ 0, & \text{else} \end{cases}, \quad \text{Age}_m = \begin{cases} 1, & \text{if middle} \\ 0, & \text{else} \end{cases}.$$

So category “Old” is the baseline level. Consider prediction of AmountSpent using Salary and Age

$$\text{AmountSpent} = \beta_0 + \beta_{\text{Salary}} \times \text{Salary} + \beta_{\text{young}} \times \text{Age}_y + \beta_{\text{middle}} \times \text{Age}_m + \epsilon$$

Linear regression model:

```
AmountSpent ~ 1 + Salary + Age_y + Age_m
```

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	171.62	69.244	2.4784	0.013361
Salary	0.022363	0.00089485	24.991	2.1484e-107
Age_y	-232.8	67.197	-3.4644	0.00055401
Age_m	-280.88	57.926	-4.8489	1.4397e-06

Spending data example

- Given the same salary, on average, a young customer spends \$232.8 less than an old customer
- Given the same salary, on average, a middle-age customer spends \$280.88 less than an old customer

How do we compare spending of young vs middle-age customers?

$$\mathbb{E}(\text{AmountSpent}|\text{young}) = \beta_0 + \beta_{\text{Salary}} \times \text{Salary} + \beta_{\text{young}}$$

$$\mathbb{E}(\text{AmountSpent}|\text{middle}) = \beta_0 + \beta_{\text{Salary}} \times \text{Salary} + \beta_{\text{middle}}$$

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Given the same salary, on average, a young customer spends \$48.08 more than a middle-age customer

Alternatively, use category “Middle” as the baseline level.

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Spending data example

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So category “Middle” is the baseline level, the other levels are compared against this level.

Linear regression model:

`AmountSpent ~ 1 + Salary + Age_y + Age_o`

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-109.26	71.159	-1.5355	0.12499
Salary	0.022363	0.00089485	24.991	2.1484e-107
Age_y	48.079	63.943	0.7519	0.45229
Age_o	280.88	57.926	4.8489	1.4397e-06

- Given the same salary, on average, a young customer spends \$48.08 more than a middle-age customer
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Interaction effects

$$\text{AmountSpent} = \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Children} + \epsilon$$

Linear regression model:

AmountSpent ~ 1 + Salary + Children

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	182.22	45.343	4.0186	6.2964e-05
Salary	0.022363	0.0006635	33.704	7.7869e-167
Children	-235.62	19.327	-12.191	5.7395e-32

- The spending amount is increased by β_1 when salary is increased by 1 unit, regardless of the number of children, as long as it's fixed
- On average, customers will spend 2.24 cents more on each \$1 increase in salary, regardless of the number of children.
- This increase in shopping spending among customers who have none children is the same as among customers who have 10 children, say. Is this reasonable?

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Interaction effects

Consider the model with an interaction term between salary and children

$$\begin{aligned}\text{AmountSpent} &= \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Children} \\ &\quad + \beta_3 \times \text{Salary} * \text{Children} + \epsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \times \text{Children}) \times \text{Salary} \\ &\quad + \beta_2 \times \text{Children} + \epsilon\end{aligned}$$

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	-61.675	56.465	-1.0923	0.27498
Salary	0.026734	0.00090221	29.632	7.1927e-139
Children	7.6915	39.705	0.19371	0.84644
Salary:Children	-0.0042338	0.00060777	-6.9661	5.9142e-12

When salary is increased by \$1, the spending amount is increased by $\beta_1 + \beta_3 \times \text{Children} \approx (2.67 - 0.42 \times \text{Children})$ cents.

Interaction effects

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- When having more children, customers spend less on the products of this company.
- To maximise its profit, the company should target customers with less children

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Hierarchy principle

$$\text{AmountSpent} = \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Children} \\ + \beta_3 \times \text{Salary} * \text{Children} + \epsilon$$

β_1 and β_2 are referred to as **main effects** and β_3 the **interaction effect**.

Sometimes it is the case that the interaction effect is significant but the associated main effects are not. If we remove the main effects then the model is not interpretable.

The hierarchy principle: If we include an interaction term in a model, the associated main effects should always be included in the model as well.

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Quadratic effects

- ABS 1995 National Survey data consists of income data on 10,827 working adults aged 18-60
- Other variables include gender, age (in years) and education (tertiary, diploma and other)
- We'd like predict income from the above predictors.
- Typically, age has a quadratic effect on income: income increases when age increases to a certain point, after this point, age has a negative effect on income.

To see if age has such a quadratic effect on income, we might consider the following model

$$\begin{aligned}\text{income} = & \beta_0 + \beta_1 \times \text{female} + \beta_2 \times \text{age} + \beta_3 \times \text{age}^2 \\ & + \beta_4 \times \text{tertiary} + \beta_5 \times \text{diploma} + \epsilon\end{aligned}$$

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Quadratic effects

- ABS 1995 National Survey data consists of income data on 10,827 working adults aged 18-60
- Other variables include gender, age (in years) and education (tertiary, diploma and other)
- We'd like predict income from the above predictors.
- Typically, age has a **quadratic** effect on income: income increases when age increases to a certain point, after this point, age has a **negative** effect on income.

To see if age has such a quadratic effect on income, we might consider the following model

$$\begin{aligned}\text{income} = & \beta_0 + \beta_1 \times \text{female} + \beta_2 \times \text{age} + \beta_3 \times \text{age}^2 \\ & + \beta_4 \times \text{tertiary} + \beta_5 \times \text{diploma} + \epsilon\end{aligned}$$

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Quadratic effects

Estimated regression equation

$$\widehat{\text{income}} = -26.70 - 7.69 \times \text{female} + 1.89 \times \text{age} - 0.02 \times \text{age}^2 \\ + 13.29 \times \text{tertiary} + 6.44 \times \text{diploma}$$

$$\frac{\partial \widehat{\text{income}}}{\partial \text{age}} = 1.89 - 0.04 \text{age}$$

When $\text{age} < 1.89/0.04 = 47.25$, age has a positive effect on income.
After this turning point, age has a negative effect on income.

Non-linear effects

- Interaction and quadratic effects are popular examples of **non-linear** effects.
- A predictor X has a non-linear effect on the response Y if the conditional mean $\mathbb{E}(Y|X)$ cannot be represented or well approximated by a linear form $\beta_0 + \beta_1 X$.
- Including appropriate non-linear effect terms into the regression model often increases the prediction accuracy
- How? this requires domain-knowledge, or trial and error. Not always easy!
- Neural network and deep learning models (not covered in this unit) provide an **efficient way** to capture non-linear effects.

Review questions

- What are the two main assumptions in the linear regression model?
- Distinguish between a covariate, a predictor and a feature
- Explain the Least Squares Method for estimating the coefficients in a linear regression model.
- Explain the Maximum Likelihood Estimation method
- Write the standard errors for the estimators of the coefficients in the SLR model.

Review questions

- What is the difference between a point estimate and a confidence interval estimate?
- Write the formula for the confidence intervals of the coefficients in the SLR model.
- Explain the logic of hypothesis testing.
- How to deal with a categorical covariate?
- What is a non-linear effect?